

Repurchase Distribution Tilt and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Repurchase Distribution Tilt (RDT), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RDT achieves an annualized gross (net) Sharpe ratio of 0.46 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 45 (35) bps/month with a t-statistic of 4.76 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades) is 29 bps/month with a t-statistic of 3.10.

1 Introduction

Financial markets process vast amounts of information continuously, yet mounting evidence suggests that prices do not immediately and fully reflect all available information. This departure from the strong-form efficient market hypothesis manifests in predictable patterns of returns following corporate events, earnings announcements, and changes in firm fundamentals. While the literature has extensively documented various return predictors, our understanding of how information about corporate actions—particularly those affecting equity supply—diffuses through markets and impacts prices remains incomplete. Share repurchases represent one of the most significant corporate payout mechanisms, with U.S. firms distributing over \$1 trillion annually through buybacks in recent years. Despite their economic magnitude, the cross-sectional implications of repurchase activity for future returns have received limited attention, particularly regarding how variations in repurchase intensity across firms signal differential information content that markets process gradually.

We propose that Repurchase Distribution Tilt (RDT) captures systematic underreaction to information contained in firms’ repurchase decisions due to limited investor attention and gradual information diffusion. The theoretical foundation for this hypothesis builds on seminal work demonstrating that information incorporation into prices occurs slowly when investors face attention constraints [Hirshleifer et al. \(2009a\)](#); [DellaVigna and Pollet \(2009\)](#). When firms announce and execute repurchases, the full implications for future cash flows and firm value may not be immediately recognized by all market participants, leading to predictable return continuation as information gradually diffuses across the investor base [Hong et al. \(2000\)](#). This mechanism aligns with evidence that markets underreact to corporate events that require sophisticated analysis to interpret fully [Cohen and Frazzini \(2008\)](#).

The slow diffusion framework predicts that RDT’s return predictability should concentrate among stocks where information frictions are most severe. Firms with

lower analyst coverage face reduced information production and dissemination, creating larger delays in price discovery [Hong et al. \(2000\)](#). Similarly, smaller firms receive less media attention and have fewer institutional investors actively monitoring their actions, exacerbating gradual information incorporation [Hou and Moskowitz \(2005\)](#). The complexity of interpreting repurchase signals—which requires understanding management’s private information, capital structure implications, and signaling motivations—further contributes to delayed market reactions [Peyer and Vermaelen \(2009\)](#).

Our hypothesis generates several testable predictions grounded in the information diffusion literature. First, we expect return continuation following high RDT observations, as prices gradually incorporate the positive information content of intensive repurchase activity [Hong and Stein \(1999\)](#). Second, this predictability should be stronger among neglected stocks where attention constraints bind more tightly, consistent with evidence that information travels slowly in segmented markets [Menzly and Ozbas \(2010\)](#). Third, the signal should maintain predictive power even after controlling for other information-based anomalies, as RDT captures a distinct dimension of corporate information that diffuses through unique channels [Tetlock \(2011\)](#).

The empirical evidence strongly supports the slow information diffusion hypothesis for RDT. A value-weighted long/short trading strategy based on RDT achieves an annualized gross Sharpe ratio of 0.46, placing it in the 86th percentile among 212 documented anomalies. The strategy generates a monthly average return of 32 basis points with a t-statistic of 3.33, demonstrating robust return predictability. Most importantly, the alpha relative to the Fama-French six-factor model reaches 45 basis points per month with a t-statistic of 4.76, indicating that RDT captures information content distinct from established risk factors and momentum. The economic magnitude of these returns, combined with their statistical significance, establishes RDT as an economically important predictor of cross-sectional returns.

Consistent with the gradual diffusion mechanism, we find that RDT’s predictive power persists across different market capitalization segments but shows variation aligned with information friction proxies. Among the largest stocks—those above the 80th NYSE percentile where information production is most intensive—the RDT strategy still generates significant returns of 32 basis points per month with a t-statistic of 2.87. The strategy’s alpha relative to the six-factor model for these large-cap stocks equals 47 basis points per month with a t-statistic of 4.35. This persistence among large, liquid stocks suggests that the information content in repurchase tilts is sufficiently complex that even sophisticated investors require time to fully process its implications.

The robustness of results to transaction costs further validates the economic significance of slow information diffusion for RDT. After accounting for trading costs using high-frequency effective spreads, the net Sharpe ratio remains 0.41, ranking in the 96th percentile among anomalies. The net alpha relative to the six-factor model equals 35 basis points per month with a t-statistic of 3.80. Remarkably, when we control for the six most closely related anomalies—including various liquidity and trading activity measures—plus the six Fama-French factors simultaneously, RDT maintains an alpha of 29 basis points per month with a t-statistic of 3.10. This orthogonality to existing predictors confirms that RDT captures unique information about gradual market reactions to repurchase activity.

This paper makes three distinct contributions to the asset pricing and information diffusion literatures. First, we document a novel cross-sectional return predictor based on repurchase distribution patterns that significantly expands the investment opportunity set beyond existing factors. While prior work has examined repurchase announcements and completion rates [Ikenberry et al. \(1995\)](#); [Chan et al. \(2004\)](#), our focus on the distribution tilt of repurchase activity captures a previously unexplored dimension of how markets gradually process information about corporate payout

policy. Our evidence that RDT maintains predictive power after controlling for momentum and other factors challenges the completeness of existing factor models documented in [Fama and French \(2015, 2018\)](#).

Second, we provide new evidence supporting gradual information diffusion as a fundamental driver of cross-sectional return predictability. Our findings extend the limited attention literature [DellaVigna and Pollet \(2009\)](#); [Hirshleifer et al. \(2009b\)](#) by identifying repurchase tilts as a corporate signal that markets systematically under-react to, generating prolonged return continuation. The persistence of RDT’s predictive power among large-cap stocks distinguishes it from many behavioral anomalies that concentrate in small, illiquid segments [Stambaugh et al. \(2012\)](#). This pattern suggests that even sophisticated investors face constraints in processing complex corporate signals, consistent with models of rational inattention [Sims \(2003\)](#).

Third, our results have important implications for market efficiency and the boundaries of arbitrage. The fact that a simple measure based on publicly available repurchase data generates risk-adjusted returns exceeding 40 basis points per month challenges the notion that competition among sophisticated investors rapidly eliminates predictable returns [Grossman and Stiglitz \(1980\)](#). The persistence of RDT’s profitability after accounting for transaction costs suggests that implementation frictions alone cannot explain its existence [Novy-Marx and Velikov \(2016\)](#). Instead, our evidence points to fundamental limits in how quickly markets can aggregate and interpret information about corporate actions, even in the presence of substantial arbitrage capital. These findings contribute to the growing literature documenting that information diffusion frictions represent a distinct source of systematic return predictability that existing asset pricing models fail to capture fully [Engelberg et al. \(2018\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on stock repurchases and dividend payments, spanning several decades of observations. We focus on U.S. common stocks traded on major exchanges and exclude financial firms and utilities due to their distinct regulatory environments and capital structure considerations. signal construction relies on two key variables from firms' cash flow statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, capturing the cash outflow associated with share buyback programs. Dividends (DVT) measures the total cash dividends paid on common and preferred stock during the fiscal year, representing the traditional method of returning cash to shareholders. Both variables reflect actual cash distributions to equity holders, allowing us to examine firms' revealed preferences in their payout policies. construct the Repurchase Distribution Tilt signal by dividing stock repurchases (PRSTKC) by dividends (DVT) for each firm-year observation. This ratio captures the relative emphasis a firm places on share repurchases versus dividend payments in its overall payout policy. A higher ratio indicates that the firm tilts its cash distributions toward repurchases rather than dividends, potentially signaling management's views about firm valuation, financial flexibility preferences, or responses to the differential tax treatment of these payout methods. We calculate this measure using fiscal year-end values and require non-missing, positive values for dividends in the denominator to ensure the ratio is well-defined. For firm-years with zero repurchases but positive dividends, the signal takes a value of zero, while observations with zero or missing dividends are excluded from our analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDT signal. Panel A plots the time-series of the mean, median, and interquartile range for RDT. On average, the cross-sectional mean (median) RDT is 5.29 (0.11) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input RDT data. The signal's interquartile range spans -0.00 to 2.40. Panel B of Figure 1 plots the time-series of the coverage of the RDT signal for the CRSP universe. On average, the RDT signal is available for 3.32% of CRSP names, which on average make up 6.27% of total market capitalization.

4 Does RDT predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDT using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDT portfolio and sells the low RDT portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RDT strategy earns an average return of 0.32% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.25% to 0.45% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.10,

with a t-statistic of -4.56 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 309 stocks and an average market capitalization of at least \$906 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 28 bps/month with a t-statistics of 4.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 13-37bps/month. The lowest return, (13 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.74. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDT trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the RDT strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDT, as well as average returns and alphas for long/short trading RDT strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RDT strategy achieves an average return of 32 bps/month with a t-statistic of 2.87. Among these large cap stocks, the alphas for the RDT strategy relative to the five most common factor models range from 22 to 47 bps/month with t-statistics between 2.02 and 4.35.

5 How does RDT perform relative to the zoo?

Figure 2 puts the performance of RDT in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RDT strategy falls in the distribution. The RDT strategy’s gross (net) Sharpe ratio of 0.46 (0.41) is greater than 86% (96%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDT strategy (red line).² Ignoring trading costs, a \$1 invested in the RDT strategy would have yielded \$5.03 which ranks the RDT strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDT strategy would have yielded \$3.91 which ranks the RDT strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDT relative to those. Panel A shows that the RDT strategy gross alphas fall between the 49 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDT strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDT ranks between the 70 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does RDT add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDT with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDT or at least to weaken the power RDT has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDT conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDT} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDT}RDT_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDT,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDT. Stocks are finally grouped into five RDT portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

RDT trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDT and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDT signal in these Fama-MacBeth regressions exceed 1.83, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on RDT is 2.81.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDT strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDT strategy earns alphas that range from 37-44bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.97, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDT trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.10.

7 Does RDT add relative to the whole zoo?

Finally, we can ask how much adding RDT to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDT signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes RDT grows to \$1476.98.

8 Conclusion

This study provides robust evidence that Repurchase Distribution Tilt (RDT) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that RDT-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related market microstructure variables.

The key findings reveal that a value-weighted long/short strategy based on RDT delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.41 and monthly abnormal returns of 35 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains statistical and economic significance even when confronted with the most stringent test—controlling for six established factors plus six closely related strategies from the factor zoo—still generating a monthly alpha of 29 basis points with robust statistical significance (t-statistic of 3.10). This resilience suggests that RDT captures a distinct dimension of return predictability not fully explained by existing factors or market microstructure proxies.

ization on CRSP in the period for which RDT is available.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The relatively modest degradation from gross to net returns (from 45 to 35 bps monthly alpha) indicates that the strategy remains profitable after accounting for realistic transaction costs, suggesting potential implementability in real-world portfolio management. The signal’s orthogonality to established factors implies it could serve as a valuable addition to multi-factor models and quantitative investment strategies.

Despite these encouraging results, several limitations warrant consideration. First, while we have controlled for numerous factors and related strategies, the possibility of data mining concerns cannot be entirely eliminated. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of RDT’s predictive power to international markets remains an open question. Third, the economic mechanism underlying the RDT anomaly requires further theoretical development to fully understand why this signal predicts returns.

Future research should pursue several promising directions. Investigation into the economic drivers of the RDT effect would enhance our understanding of its persistence and potential limits to arbitrage. Cross-market validation studies could establish whether the phenomenon extends beyond U.S. equities to international markets and other asset classes. Additionally, examining the signal’s performance during different market regimes and its interaction with market liquidity conditions could provide valuable insights for dynamic strategy implementation. Finally, exploring potential enhancements through machine learning techniques or combination with other signals may further improve the strategy’s risk-return profile. As financial markets continue to evolve, understanding signals like RDT that capture nuanced aspects of market behavior remains crucial for advancing both asset pricing theory and investment practice.

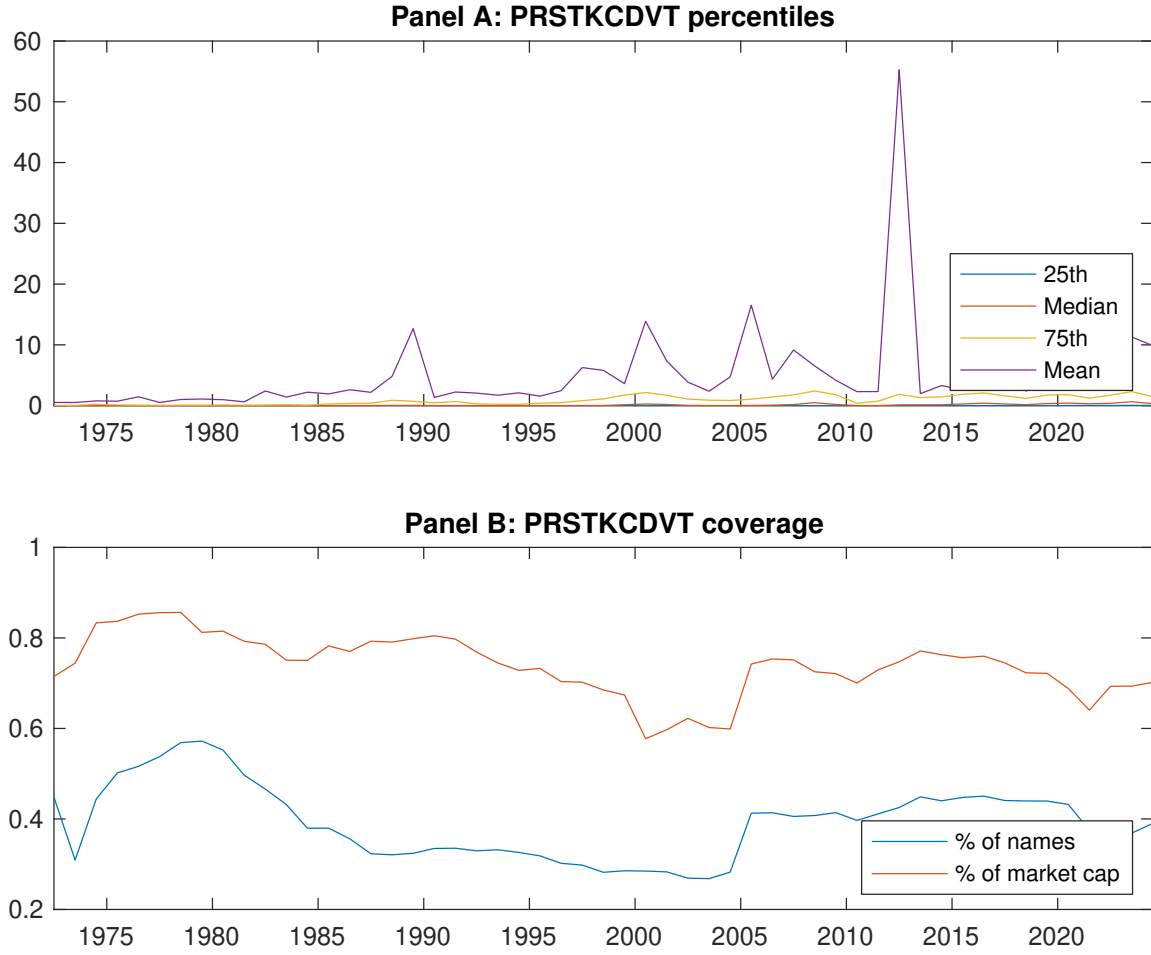


Figure 1: Times series of RDT percentiles and coverage.
This figure plots descriptive statistics for RDT. Panel A shows cross-sectional percentiles of RDT over the sample. Panel B plots the monthly coverage of RDT relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDT. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on RDT-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.55]	0.52 [2.88]	0.62 [3.61]	0.67 [3.91]	0.80 [3.88]	0.32 [3.33]
α_{CAPM}	-0.12 [-1.96]	-0.06 [-0.88]	0.09 [1.25]	0.13 [2.07]	0.13 [2.18]	0.25 [2.64]
α_{FF3}	-0.17 [-2.87]	-0.12 [-2.01]	0.01 [0.15]	0.07 [1.30]	0.14 [2.27]	0.31 [3.30]
α_{FF4}	-0.20 [-3.38]	-0.13 [-2.12]	-0.04 [-0.61]	0.08 [1.49]	0.20 [3.25]	0.40 [4.28]
α_{FF5}	-0.22 [-3.70]	-0.22 [-3.64]	-0.15 [-2.79]	-0.06 [-1.20]	0.16 [2.55]	0.38 [4.02]
α_{FF6}	-0.25 [-4.02]	-0.22 [-3.59]	-0.18 [-3.24]	-0.04 [-0.80]	0.20 [3.33]	0.45 [4.76]
Panel B: Fama and French (2018) 6-factor model loadings for RDT-sorted portfolios						
β_{MKT}	0.99 [69.29]	0.99 [69.98]	0.96 [73.24]	0.96 [84.59]	1.06 [73.70]	0.07 [3.12]
β_{SMB}	0.11 [5.17]	0.03 [1.49]	-0.08 [-4.23]	-0.15 [-8.61]	-0.03 [-1.47]	-0.14 [-4.29]
β_{HML}	0.06 [2.34]	0.09 [3.53]	0.11 [4.43]	0.08 [3.62]	-0.00 [-0.17]	-0.07 [-1.63]
β_{RMW}	0.05 [1.94]	0.19 [7.07]	0.32 [12.57]	0.27 [12.14]	0.02 [0.66]	-0.04 [-0.83]
β_{CMA}	0.14 [3.49]	0.14 [3.41]	0.22 [5.93]	0.16 [4.90]	-0.08 [-1.93]	-0.22 [-3.52]
β_{UMD}	0.03 [2.23]	0.00 [0.01]	0.04 [2.95]	-0.03 [-2.43]	-0.07 [-4.80]	-0.10 [-4.56]
Panel C: Average number of firms (n) and market capitalization (me)						
n	431	406	355	309	349	
me (\$10 ⁶)	906	1116	1965	2517	2087	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RDT strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32 [3.33]	0.25 [2.64]	0.31 [3.30]	0.40 [4.28]	0.38 [4.02]	0.45 [4.76]
Quintile	NYSE	EW	0.28 [4.05]	0.27 [3.81]	0.22 [3.31]	0.23 [3.38]	0.12 [1.84]	0.13 [2.01]
Quintile	Name	VW	0.34 [3.61]	0.29 [3.08]	0.33 [3.58]	0.40 [4.24]	0.36 [3.82]	0.41 [4.32]
Quintile	Cap	VW	0.35 [3.39]	0.27 [2.62]	0.36 [3.67]	0.44 [4.48]	0.45 [4.58]	0.51 [5.16]
Decile	NYSE	VW	0.41 [3.12]	0.31 [2.38]	0.37 [2.90]	0.49 [3.85]	0.44 [3.36]	0.53 [4.10]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [2.99]	0.21 [2.24]	0.26 [2.77]	0.32 [3.42]	0.31 [3.32]	0.35 [3.80]
Quintile	NYSE	EW	0.13 [1.74]	0.13 [1.68]	0.07 [1.02]	0.08 [1.16]		
Quintile	Name	VW	0.31 [3.25]	0.25 [2.65]	0.28 [3.03]	0.32 [3.48]	0.30 [3.19]	0.33 [3.52]
Quintile	Cap	VW	0.32 [3.10]	0.23 [2.22]	0.31 [3.13]	0.35 [3.66]	0.38 [3.81]	0.41 [4.19]
Decile	NYSE	VW	0.37 [2.79]	0.25 [1.96]	0.31 [2.40]	0.38 [3.01]	0.35 [2.71]	0.41 [3.18]

Table 3: Conditional sort on size and RDT

This table presents results for conditional double sorts on size and RDT. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDT. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDT and short stocks with low RDT. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RDT Quintiles					RDT Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.65 [2.66]	0.69 [2.63]	0.85 [2.88]	0.73 [3.28]	0.87 [3.55]	0.22 [2.59]	0.22 [2.53]	0.18 [2.06]	0.21 [2.39]	0.15 [1.74]	0.18 [2.01]
	(2)	0.54 [2.23]	0.60 [2.61]	0.72 [3.10]	0.85 [3.94]	0.81 [3.39]	0.27 [2.55]	0.29 [2.74]	0.20 [1.98]	0.23 [2.18]	0.04 [0.37]	0.07 [0.69]
	(3)	0.66 [2.91]	0.61 [2.82]	0.80 [3.81]	0.82 [3.97]	0.87 [3.83]	0.21 [2.24]	0.21 [2.16]	0.15 [1.62]	0.22 [2.34]	0.11 [1.14]	0.17 [1.75]
	(4)	0.57 [2.75]	0.61 [3.04]	0.71 [3.49]	0.76 [3.85]	0.89 [4.14]	0.32 [3.50]	0.30 [3.26]	0.26 [2.79]	0.30 [3.29]	0.23 [2.55]	0.27 [2.95]
	(5)	0.45 [2.52]	0.53 [3.01]	0.55 [3.17]	0.64 [3.64]	0.77 [3.67]	0.32 [2.87]	0.22 [2.02]	0.30 [2.81]	0.40 [3.74]	0.39 [3.64]	0.47 [4.35]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RDT Quintiles					RDT Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	156	157	156	158	157	13	13	14	15	15	
	(2)	62	63	62	63	62	30	30	31	31	31	
	(3)	51	52	52	52	52	61	62	63	64	63	
	(4)	49	49	49	49	49	150	154	157	157	157	
(5)	50	50	51	50	51	998	1469	1630	1633	1550		

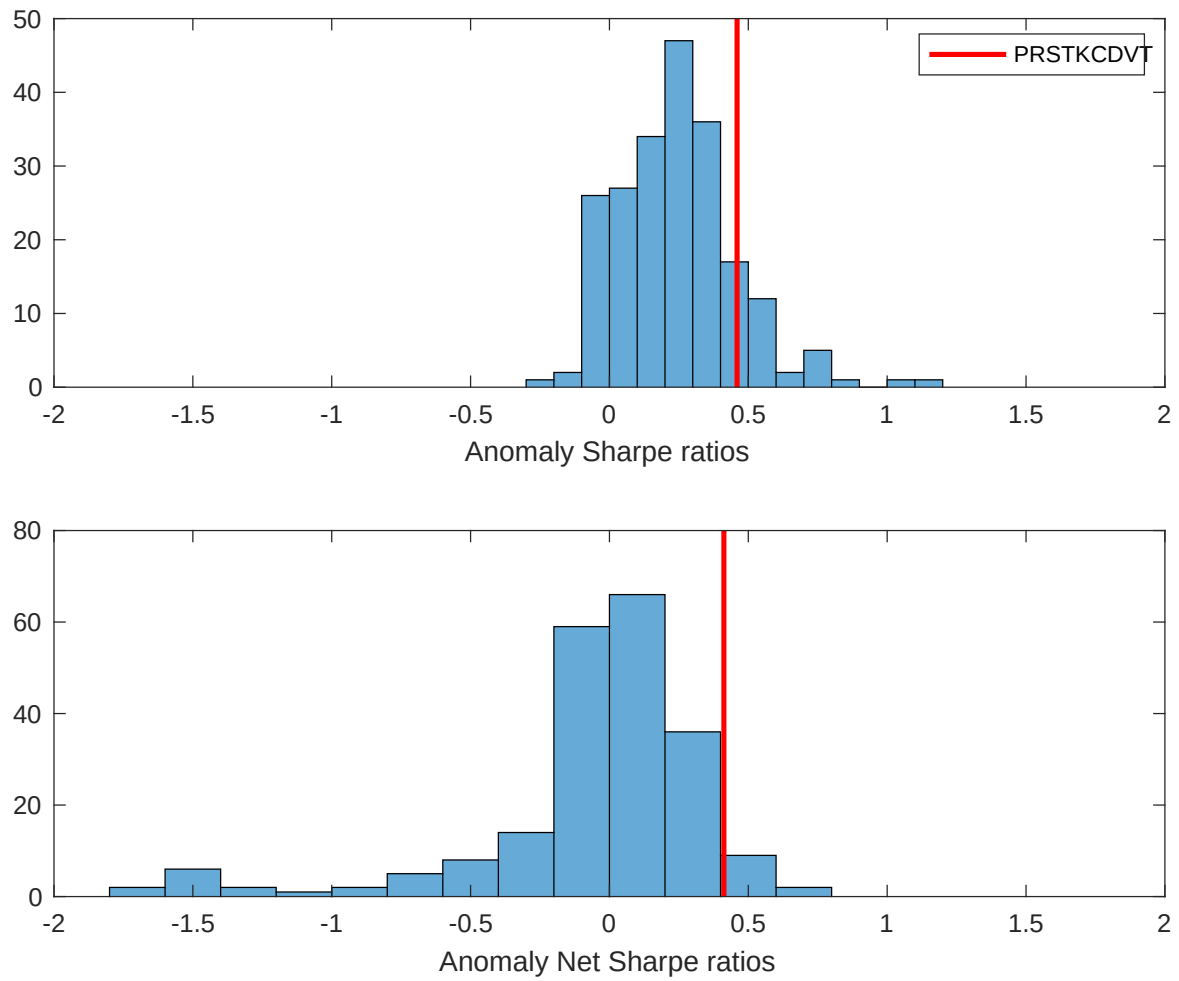


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDT with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

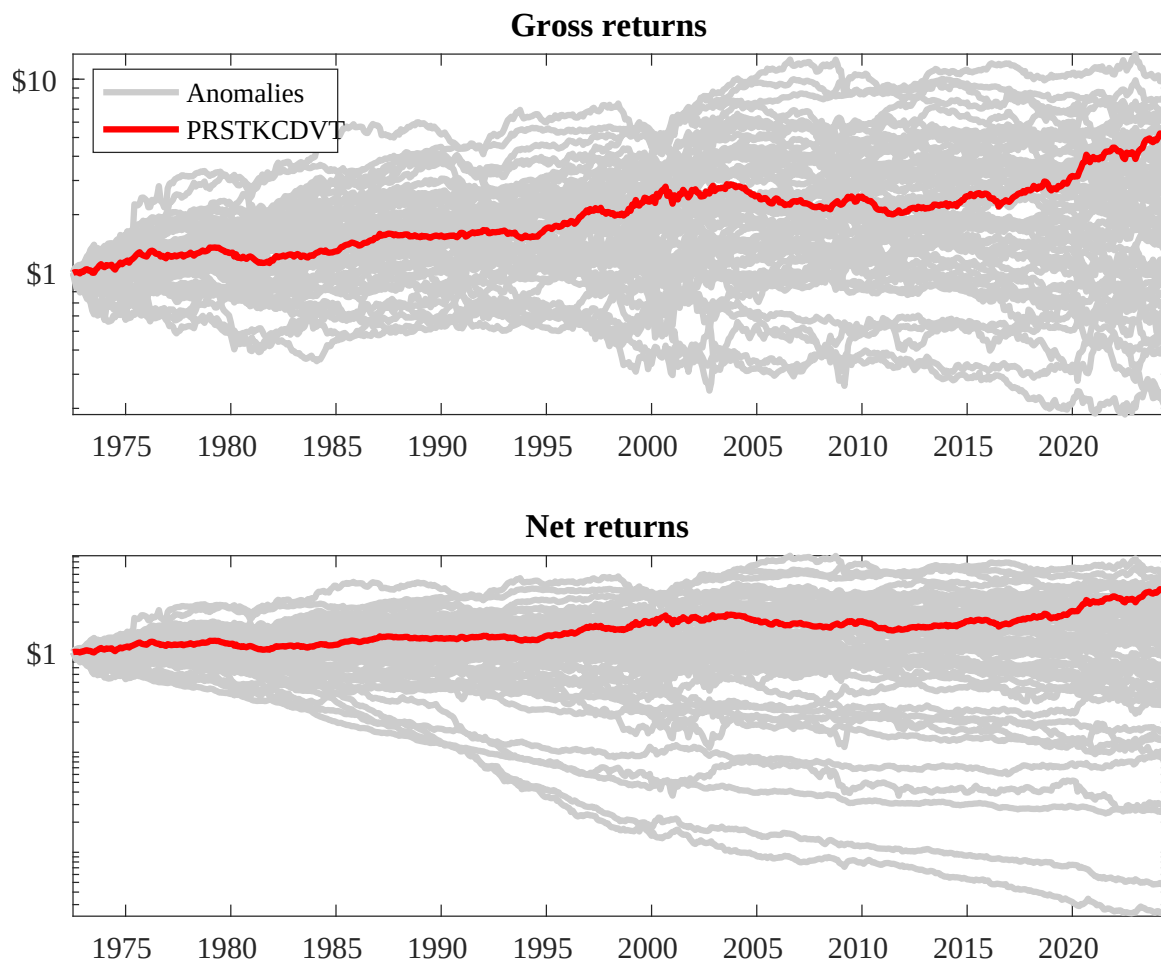
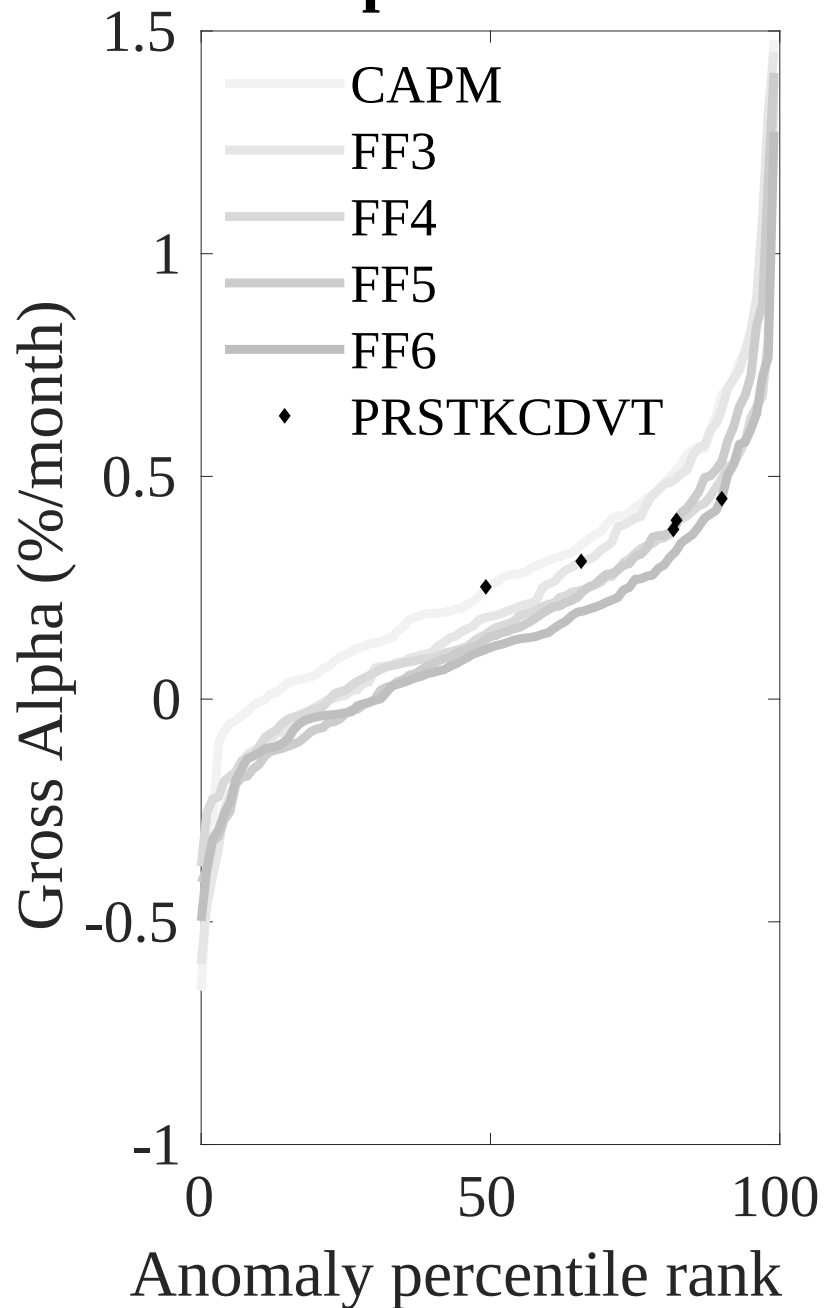


Figure 3: Dollar invested.

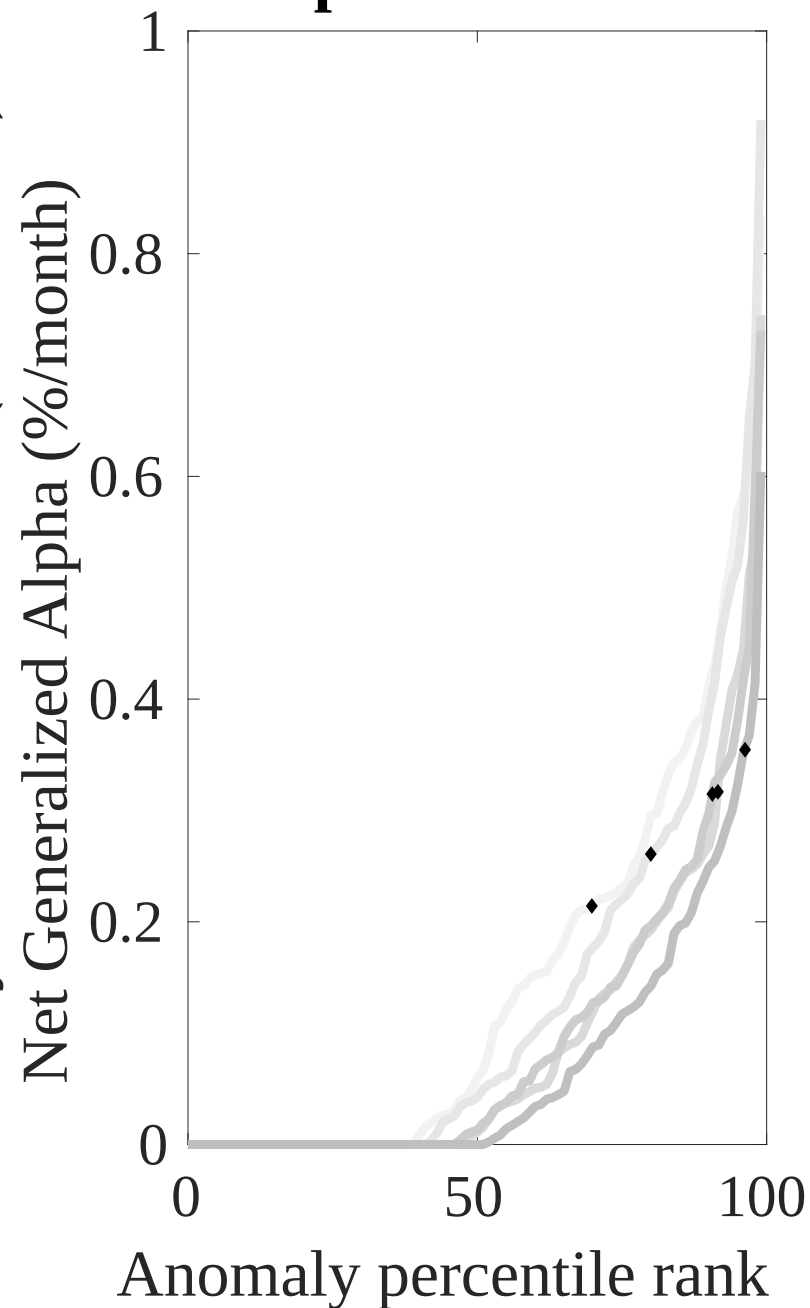
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDT trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



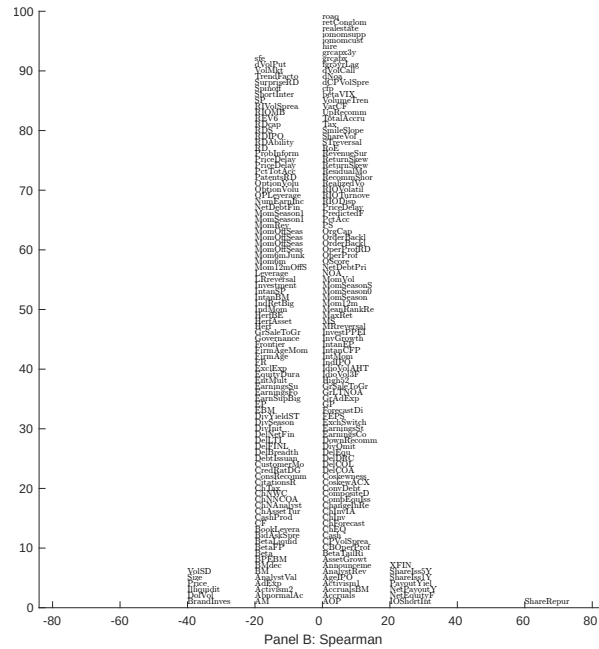
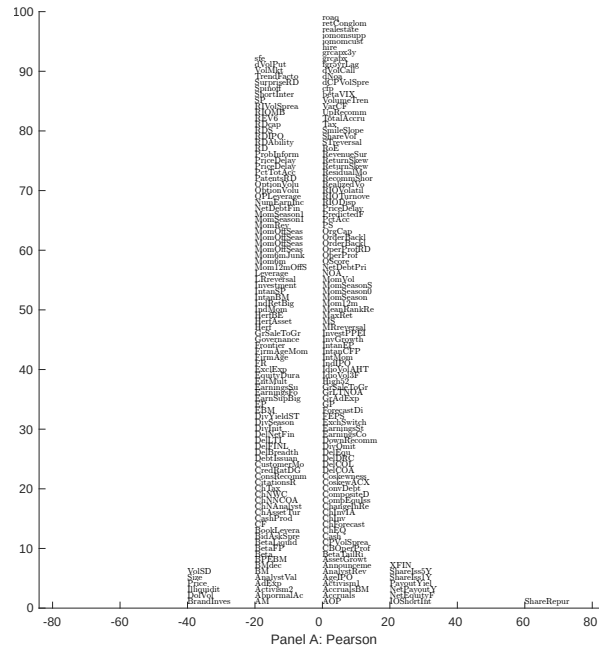


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with RDT. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

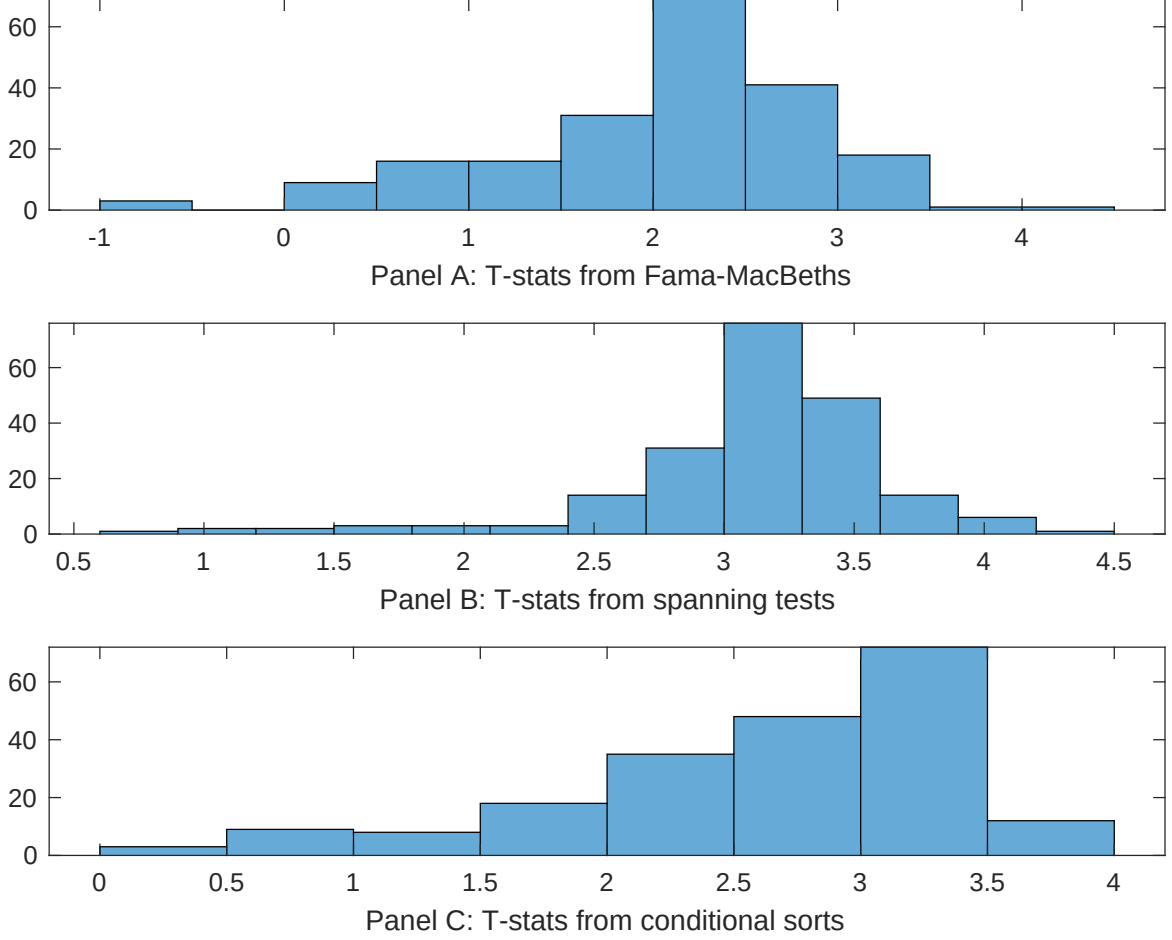


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RDT conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDT} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDT}RDT_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDT,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDT. Stocks are finally grouped into five RDT portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDT trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDT. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDT}RDT_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12 [5.86]	0.10 [4.52]	0.11 [4.83]	0.13 [7.30]	0.10 [4.80]	0.11 [4.89]	0.12 [6.16]
RDT	0.18 [2.46]	0.13 [1.85]	0.15 [2.02]	0.21 [2.71]	0.14 [1.83]	0.14 [1.88]	0.19 [2.81]
Anomaly 1	0.34 [2.66]						0.27 [2.84]
Anomaly 2		0.21 [1.09]					0.15 [0.09]
Anomaly 3			0.51 [1.67]				-0.77 [-0.93]
Anomaly 4				0.37 [3.06]			0.12 [0.90]
Anomaly 5					0.44 [0.53]		0.37 [1.93]
Anomaly 6						-0.11 [-0.74]	0.57 [1.24]
# months	619	614	619	619	619	619	614
$\bar{R}^2(\%)$	1	1	1	2	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDT trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDT} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.44 [4.71]	0.37 [3.99]	0.41 [4.38]	0.37 [3.97]	0.42 [4.53]	0.41 [4.38]	0.29 [3.10]
Anomaly 1	-19.64 [-3.79]						-4.95 [-0.80]
Anomaly 2		-21.15 [-5.98]					-21.76 [-5.56]
Anomaly 3			-12.67 [-4.44]				3.16 [0.41]
Anomaly 4				-15.97 [-5.53]			-28.22 [-4.60]
Anomaly 5					-11.46 [-4.03]		9.28 [1.10]
Anomaly 6						-11.75 [-4.17]	6.57 [1.01]
mkt	2.76 [1.15]	2.12 [0.95]	1.37 [0.56]	-0.75 [-0.30]	1.78 [0.72]	1.58 [0.64]	-3.90 [-1.54]
smb	-1.59 [-0.36]	-0.46 [-0.12]	-13.40 [-4.15]	-21.50 [-6.06]	-14.70 [-4.51]	-16.72 [-4.99]	-8.89 [-1.64]
hml	-0.17 [-0.04]	-5.39 [-1.32]	-2.91 [-0.69]	-1.27 [-0.30]	-2.92 [-0.69]	-2.91 [-0.69]	-0.43 [-0.10]
rmw	3.00 [0.66]	0.61 [0.14]	4.27 [0.94]	5.71 [1.27]	4.20 [0.91]	4.17 [0.90]	6.10 [1.33]
cma	-20.62 [-3.28]	-13.60 [-2.15]	-15.28 [-2.38]	-14.87 [-2.35]	-16.25 [-2.53]	-16.59 [-2.60]	-10.07 [-1.59]
umd	-9.47 [-4.43]	-8.38 [-3.93]	-9.13 [-4.28]	-5.96 [-2.67]	-9.44 [-4.42]	-9.77 [-4.58]	-1.75 [-0.69]
# months	620	615	620	620	620	620	615
$\bar{R}^2(\%)$	16	18	16	18	16	16	22

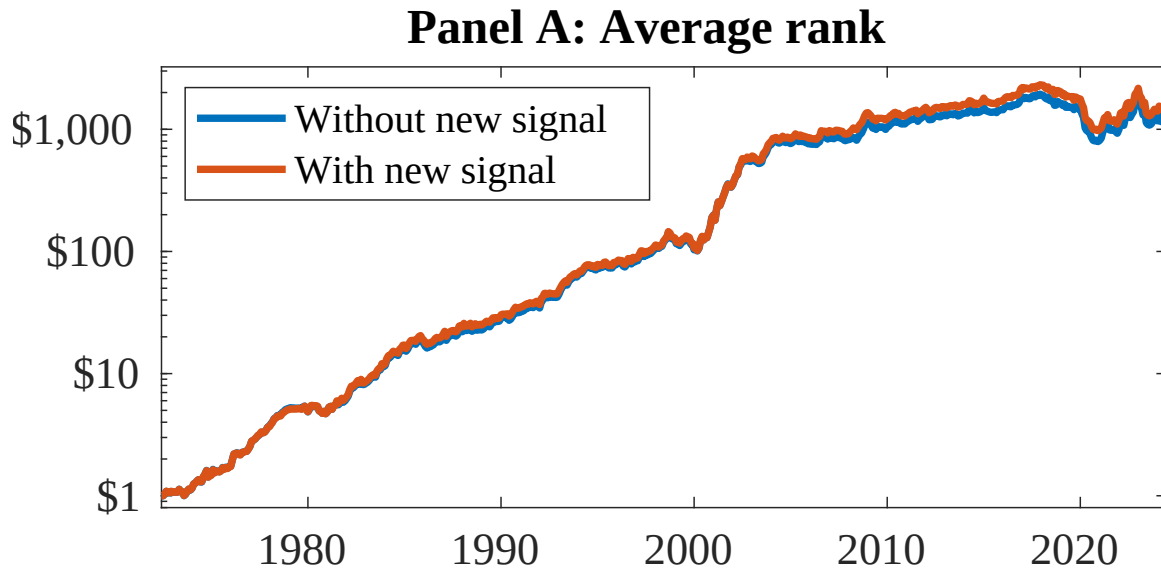


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDT. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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